

- Q.26 How does the Dewas system differ from the Gabler system?
- Q.27 What are the main advantages of air-jet looms?
- Q.28 How does air-jet weft insertion differ from shuttle weft insertion?
- Q.29 Explain the working principle of a water-jet looms?
- Q.30 What types of yarns are best suited for water-jet weaving?
- Q.31 List two disadvantages of water-jet looms.
- Q.32 Explain why air-jet looms are faster than shuttle looms.
- Q.33 What is the role of the reed in the weft insertion process?
- Q.34 What is the significance of using multiple projectiles in projectile weaving?
- Q.35 List three advantages of rapier looms.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the concept of unconventional weaving machines. Discuss their importance in modern textile production and compare them with traditional shuttle looms.
- Q.37 Describe the working principle of a projectile loom. Explain the sequence of weft insertion, the role of the gripper, and the return mechanism.
- Q.38 Compare water-jet looms with air-jet looms. Explain their differences in terms of energy usage, fabric types.

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5th Sem / Text. Design Subject:- Fabric Manufacturing - III

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is NOT an unconventional weft insertion method?
- a) Projectile picking b) Shuttle picking
c) Rapier picking d) Air-jet picking
- Q.2 Which advantage is associated with conventional weaving machines over shuttle looms?
- a) Higher noise levels
b) Slower production rate
c) Lower wear and tear
d) Increased shuttle vibration
- Q.3 Which device carries the weft across the shed in a projectile loom?
- a) Shuttle b) Rapier
c) Gripper d) Jet nozzle
- Q.4 What happens after the projectile inserts the weft?
- a) The weft is beaten into place by the reed
b) The rapier takes over
c) The warp is advanced
d) The shuttle returns with the weft

- Q.5 What is the main advantage of a rapier loom over a projectile loom?
- Faster speed
 - Better suitability for intricate patterns
 - Lower cost
 - Uses a shuttle for insertion
- Q.6 Which system uses a single rapier to insert the weft?
- Dewas system
 - Shuttle system
 - Gabler system
 - Water-jet system
- Q.7 Which component is responsible for creating the air pressure in an air-jet loom?
- Feeder mechanism
 - Nozzle
 - Reed
 - Projectile
- Q.8 Which type of yarn is best suited for air-jet weaving?
- Heavy cotton yarn
 - Fine synthetic yarn
 - Woolen yarn
 - Coarse jute yarn
- Q.9 Which type of yarn is best suited for water-jet looms?
- Cotton
 - Wool
 - Synthetic yarns
 - Linen
- Q.10 What is a major limitation of water-jet looms?
- Low weaving speed
 - High power consumption
 - Cannot handle hydrophilic fibers like cotton and wool
 - High maintenance requirement

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ is a type of unconventional picking mechanism.
- Q.12 Function of a weft accumulator is _____
- Q.13 In a projectile loom, how is the weft inserted?
- Q.14 Which type of selvedge is commonly formed in projectile looms?
- Q.15 _____ is a method of weft insertion in a rapier loom?
- Q.16 Any one advantage of a rapier loom over a projectile loom.
- Q.17 _____ medium used for weft insertion in an air-jet loom?
- Q.18 _____ type of yarn is best suited for air-jet weaving?
- Q.19 Any one disadvantage of air-jet looms.
- Q.20 What is a major limitation of water-jet looms?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are unconventional weaving machines?
- Q.22 Differentiate between conventional and tucked selvedge.
- Q.23 What is a projectile loom?
- Q.24 How is the weft inserted in a projectile loom?
- Q.25 Why are rapier looms preferred for weaving intricate designs?